Project title: Systems Analysis of Speculative Decoding on Memory-Constrained Hardware

- **Introduction:**
  - Autoregressive LLM inference is inherently sequential and limited by GPU memory bandwidth rather than compute.
  - Existing solutions such as Speculative Decoding works but they are largely analysed on large server-class GPUs (e.g. A100s/H100s).
  - This project implements Speculative Decoding from scratch with a system-level performance evaluation on memory-constrained/consumer hardware.
- **Motivation:**
  - To introduce a minimal, transparent inference engine that isolates Speculative Decoding logic from heavy, complex production frameworks.
  - To specifically expose and benchmark the hardware bottlenecks - like KV cache management - that are currently hidden by enterprise-level abstractions.
  - To bridge the gap between theoretical papers and production code, providing a clear baseline that allows for direct study and student research.

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Existing work / Literature review:

- **Foundational Algorithm [1, 2]:**
  - Introduced the **Draft-then-Verify** mechanism to break the sequential bottleneck.
  - Proved inference can be accelerated without losing model quality.
- **Hardware Bottleneck Analysis [3]:**
  - Established that Key-Value (KV) cache allocation is the primary bottleneck in autoregressive inference.
  - Provided the literature baseline for why managing KV cache rollbacks is a critical system constraint.

- **Gaps in the existing solutions/work**:
  - **"Black Box" Frameworks:** Production tools (vLLM/Hugging Face) hide architectural logic, preventing direct systems-level study.
  - **Hardware Bias:** Existing benchmarks focus on **A100/H100 clusters**, ignoring performance on consumer-grade hardware.
  - **Hidden Overheads:** The actual cost of **KV Cache rollbacks** is rarely measured or discussed in standard literature.

[1] Leviathan et al. (2023). **Fast Inference from Transformers via Speculative Decoding**. *ICML*.
[2] Chen et al. (2023). **Accelerating LLM Decoding with Speculative Decoding**. *arXiv:2302.01318*.
[3] Kwon et al. (2023). ***Efficient Memory Management for LLM Serving with PagedAttention***. *SOSP*.

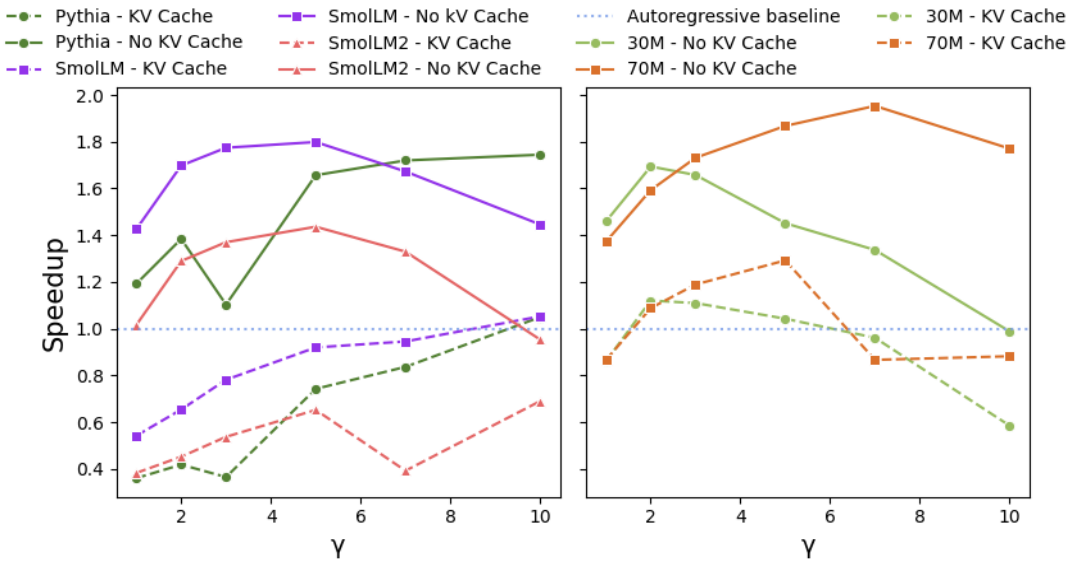
Work Done till date:

- **Details of proposed work/solutions:**

- This project addresses the latency bottleneck by implementing and analyzing a **Speculative Decoding Inference Engine.**
- **Core Mechanism:** Implements a **Draft-then-Verify** pipeline to decouple generation speed from model size.
- **Systems Focus:** Benchmarking the transition between **Compute-bound (cacheless)** and **Memory-bound (cached)** regimes. It is not designed to achieve state-of-the-art modeling performance.
- **Expected Speedup Math:** The system achieves acceleration when the expected value (E) of consecutive exact token matches $E = \sum_{i=1}^{gamma} P(match_i)$ generates tokens faster than the target model's sequential baseline.
- **Experimental Setup:**
  - **Models:** 30M/70M Draft models paired with a 150M Target model.
  - **Variables:** Controlled testing of lookahead depth (γ), context length, and caching strategies.
  - **Cross-Model Validation:** Used Pythia, SmolLM and SmolLM2 models to validate the results.

- **Result achieved:**

Method	γ	Alignment	KV Cache	TPS	Speedup
Baseline	-	-	Off	27.0	1.0x
Speculative (70M)	7	~70%	Off	52.7	1.95x
Speculative (30M)	2	~30%	Off	45.7	1.69x
Speculative (70M)	7	~70%	On	39.0	0.86x*



*Relative to Baseline (KV Cache On), TPS = 45.10. KV Cache overhead on the draft model currently exceeds speculative gains.